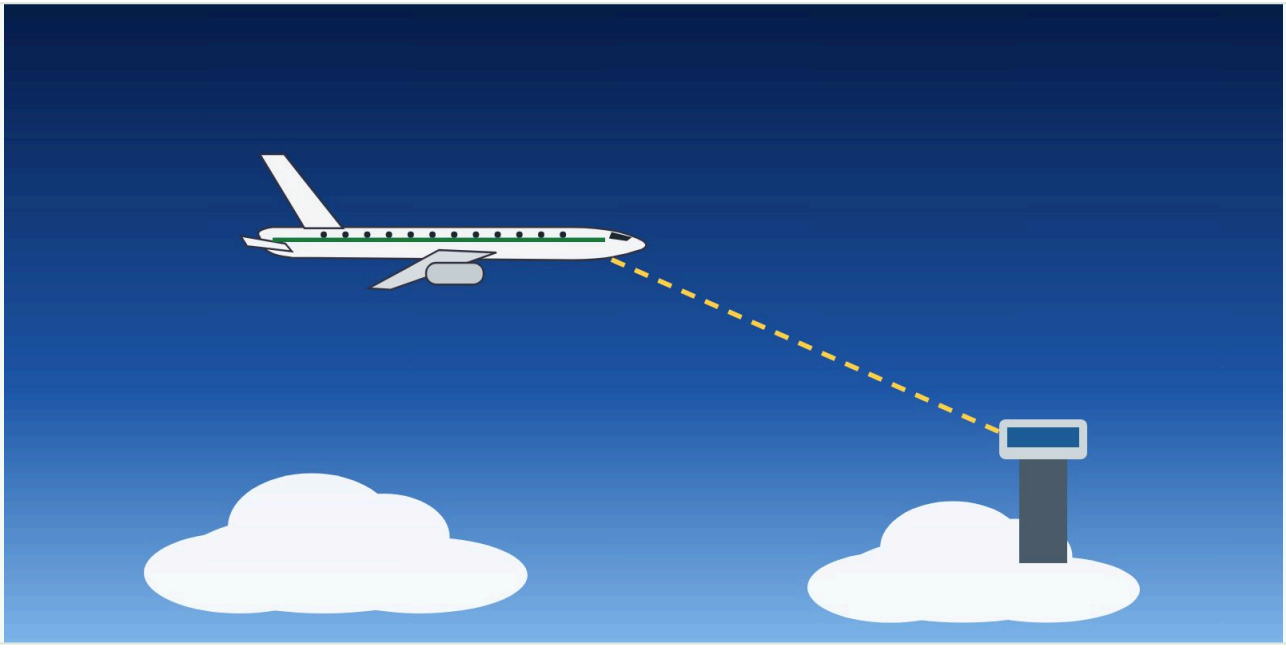
MODULE 1 · CPDLC PRINCIPLES

Lesson 01: CPDLC: Definition, References and Benefits

Lesson 1 of 6 · about 7 min

LEARNING OBJECTIVES

- Define CPDLC and describe its purpose.
- Identify the ICAO and European references.
- List the expected benefits and limitations.



CPDLC: data link messages between controller and pilot.

Definition

Controller–Pilot Data Link Communications (CPDLC) is a means of communication between controller and pilot using data link for ATC communications (ICAO Doc 4444, PANS-ATM). It is a two-way system by which controllers can send **non-urgent, strategic messages** to an aircraft as an alternative to voice. The message is displayed on a flight deck display.

- Controllers can issue clearances (levels, routes and deviations, speeds), frequency assignments and requests for information.
- Pilots can respond to messages, request and receive clearances and information, and send reports.
- A **free text** capability exchanges information that does not fit the standard formats.

References

Reference	Content
ICAO Annex 10, Volume III	Communication systems, including data link

ICAO Doc 4444 (PANS-ATM)	CPDLC procedures and the CPDLC message set
ICAO Doc 10037 — GOLD (Global Operational Data Link) Manual	The global operational guidance for flight crews, operators and ATS on data link operations
Commission Implementing Regulation (EU) 2023/1770	Current EU requirements for aircraft equipment and data link services in Single European Sky airspace. It **repealed** the original Data Link Services Implementing Rule, Regulation (EC) No 29/2009.
Aircraft manuals	FCOM / AFM limitations and the CMU operating procedures for your type

A short history in Europe

The Data Link Services Implementing Rule (Regulation (EC) No 29/2009) mandated CPDLC above FL285 in European airspace. Its implementation was affected by technical problems — random disconnections called **provider aborts**. An EASA investigation found that they came from a combination of factors linked to the radio-frequency environment and the single-frequency data link infrastructure, and recommended a multi-frequency infrastructure. The dates of application were therefore deferred (Regulation (EU) 2015/310 and later amendments). In 2023 the requirements were consolidated in Regulation (EU) 2023/1770.

Benefits

- Less voice traffic on the frequency and increased sector capacity;
- more pilot requests handled at the same time;
- fewer misunderstandings — for example call sign confusion;
- safer frequency changes and fewer loss-of-communication events.

LIMITATION

CPDLC latency is much longer than voice — a response can take up to several minutes. Use it only for **non-time-critical** communications. Check your aircraft limitations: on the B737 with protected-mode CPDLC (PM-CPDLC), the installation is intended for use in the **cruise** phase and for non-critical messages only.

KEY TAKEAWAYS

- CPDLC: data link ATC messages for non-urgent, strategic communications.
- References: Annex 10 Vol III, PANS-ATM, ICAO GOLD (Doc 10037), EU Regulation 2023/1770.
- Benefits: less frequency congestion, fewer misunderstandings; limitation: latency — not for time-critical messages.

KNOWLEDGE CHECK

Which regulation now contains the EU data link requirements, replacing Regulation (EC) No 29/2009?

A. Regulation (EU) 965/2012

B. Commission Implementing Regulation (EU) 2023/1770

C. Regulation (EU) 2015/340

D. ICAO Doc 4444

Answer: B. Regulation (EU) 2023/1770 repealed Regulation (EC) No 29/2009.

Training material only. It does not replace the Aircraft Flight Manual, FCOM, QRH, MEL or the Domestic Airlines Operations Manual, which always take precedence.